

Rational functions

A rational function is a quotient of polynomials on inputs where its denominator is nonzero.

$$f(x) = \frac{x + 1}{x - 3}$$

What is its domain?

Factoring the denominator

$$f(x) = \frac{x + 2}{x^2 - 9}$$

Which inputs are excluded from the domain?

Cancellation and the domain

$$f(x) = \frac{x^2 - 4}{x - 2}$$

What restriction remains after canceling a common factor?

A hole

A canceled denominator factor can leave a missing point in the original graph.

$$f(x) = \frac{x^2 - 4}{x - 2}$$

What are the coordinates of the hole?

Vertical asymptotes

A vertical asymptote occurs where outputs become unbounded from at least one side.

$$f(x) = \frac{1}{x - 3}$$

What vertical line is its asymptote?

A hole and an asymptote

$$f(x) = \frac{x^2 - 1}{x^2 - x - 2}$$

Which excluded input creates a hole, and which creates a vertical asymptote?

One-sided arrow notation

$$x \rightarrow 2^-$$

From which side is x approaching 2?

One-sided behavior

$$f(x) = \frac{1}{x - 2}$$

What happens to $f(x)$ as x approaches 2 from each side?

An even denominator power

$$f(x) = \frac{1}{(x + 1)^2}$$

How does the output behave on the two sides of $x = -1$?

Horizontal asymptotes

A horizontal asymptote describes a graph's behavior as inputs become unbounded.

$$f(x) = \frac{3x + 2}{x + 5}$$

What horizontal line describes its end behavior?

Lower numerator degree

$$f(x) = \frac{4x - 1}{x^2 + 1}$$

What horizontal asymptote does the function have?

Equal degrees

$$f(x) = \frac{2x^2 + 1}{5x^2 - 3}$$

What is the horizontal asymptote?

A slant asymptote

$$f(x) = \frac{x^2 + 1}{x - 1}$$

What slant asymptote does polynomial division reveal?

Crossing a horizontal asymptote

$$f(x) = \frac{x^2 + x}{x^2 + 1}$$

Where does the graph cross its horizontal asymptote $y = 1$?

Finding intercepts

$$f(x) = \frac{x - 2}{x + 3}$$

What are the horizontal and vertical intercepts?

A canceled numerator zero

$$f(x) = \frac{(x - 1)(x + 2)}{x - 1}$$

Is $x = 1$ a zero of the original function?

Signs between critical inputs

$$f(x) = \frac{x - 2}{x + 1}$$

On what intervals is $f(x)$ positive?

A rational inequality

$$\frac{x - 2}{x + 1} \leq 0$$

Which inputs satisfy this inequality?

Sketching from key features

$$f(x) = \frac{x + 1}{x - 2}$$

What sketch matches its domain, intercepts, asymptotes, and signs?

A graph with a hole

$$f(x) = \frac{x^2 - 1}{x^2 + x - 2}$$

What sketch includes both the hole and the vertical asymptote?

Average cost

A workshop has \$300 fixed cost and \$12 cost per item. Let n be a positive integer.

What rational function gives average cost per item?

Interpreting an asymptote

Average cost in dollars for n items is:

$$A(n) = 12 + \frac{300}{n}, \quad n > 0$$

What does the horizontal asymptote mean in this model?

Combined work rates

One machine finishes a job in x hours and another in $x + 2$ hours. Assume constant rates and $x > 0$.

What rational function gives their combined job rate?

A domain-aware intersection

$$f(x) = \frac{2}{x-1}, \quad g(x) = x$$

At which points do these graphs intersect?